MODEL X 2.0

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PHASE 1 · PROJECT PROPOSAL

Team name EchoBinary

IEEE Computational Intelligence Society Student Chapter, IIT

Choose one domain

☐	**01**	Cost of living and everyday economics
☐	**02**	Climate, environment and disaster risk
[x]	**03**	Health and wellbeing - non-clinical only: no diagnosis, no treatment advice

1. Problem statement and definition

Dengue remains a major public-health problem in Sri Lanka. Current dashboards and surveillance reports can show where case numbers are already high, but they are less useful for identifying areas that are likely to become hotspots next.

This creates a reactive decision-making problem. Public-health teams have limited resources for inspections, vector-control activities, cleanup programmes, and awareness campaigns. If an area is recognised only after cases have already increased significantly, the opportunity for earlier intervention is reduced.

DengueAhead addresses the following problem:

How can public-health officers identify geographic areas that are likely to become unusually high-risk for dengue one to two weeks in advance?

The system focuses on population-level forecasting only. It does not diagnose individuals or provide treatment advice.

2. The user

The primary user is a **regional or MOH-level public-health officer responsible for prioritising dengue-prevention activities**.

Their recurring decision is: **Which areas should receive extra monitoring or preventive attention during the next one to two weeks?**

Today, this decision is based on recent case reports, known hotspots, field observations, weather conditions, and professional judgement. DengueAhead adds a forward-looking view by ranking areas according to their predicted near-term dengue risk.

Instead of only showing: **Area A — currently 80 cases**

the system may highlight: **Area B — currently 35 cases, but high probability of becoming an emerging hotspot next week.**

3. Literature review

Existing dengue surveillance systems are useful for monitoring current cases and identifying areas that already have high incidence. Research has also shown that dengue forecasting can use historical case counts, seasonality, and environmental variables such as rainfall and temperature.

Machine-learning and time-series methods have been applied to dengue prediction, but many systems focus mainly on forecasting overall case counts rather than helping a specific user prioritise future geographic hotspots. The gap addressed by DengueAhead is:

A practical, explainable system that identifies emerging geographic dengue risk before it becomes obvious from current case counts.

The system also differs from a normal dashboard by displaying uncertainty, data freshness and reasons behind each prediction.

4. Proposed solution

DengueAhead will be a web-based early-warning dashboard.

For each supported geographic area, the system will estimate:

- Predicted dengue activity for the next week.
- Probability of becoming high-risk.
- Difference from the normal seasonal pattern.
- Main factors contributing to the prediction.
- Confidence and data freshness.

Areas will be grouped into categories such as:

- **Stable**
- **Emerging**
- **Persistent**
- **Cooling**

The main value is the **Emerging** category.

Example:

MOH Area X

Current cases: 24
Forecast next week: 39–50
Seasonal expectation: 16–25
Probability of high-risk activity: 82%
Status: **Emerging Hotspot**

Main signals:

- Cases have increased for three consecutive weeks.
- Rainfall two weeks earlier was above normal.
- Nearby areas are also showing rising cases.

This is information the user could not easily derive manually across many areas and several data sources.

5. Data

5.1 Source and access

The main dengue data source will be the **National Dengue Control Unit, Sri Lanka**.

Sources include:

https://www.dengue.health.gov.lk/

https://www.dengue.health.gov.lk/weekly-report/

Historical dengue data may also be obtained from publicly available datasets derived from official NDCU reports.

Weather data will be obtained from an open weather source such as Open-Meteo:

https://open-meteo.com/

Population and geographic information may be sourced from the Sri Lanka Department of Census and Statistics and public geographic datasets. The project will use aggregated geographic data only and will not collect identifiable patient information.

5.2 Extraction method

Data will be collected using Python scripts through:

- Direct file downloads.
- HTML/PDF extraction where required.
- Weather APIs.
- Public historical datasets.

Cleaning will include:

- Standardising area names.
- Standardising weeks and dates.
- Removing duplicate records.
- Detecting missing weeks.
- Checking invalid values.

- Matching weather data to geographic areas.

If MOH-level historical data is incomplete, district-level forecasting will be used instead.

5.3 Temporality

Historical data will be used for training and backtesting. Recent dengue reports and weather data will be used for current predictions. The system will display when each source was last updated. If a source becomes stale, the dashboard will warn the user. If the data is too incomplete to support a reliable forecast, the system will withhold the prediction instead of presenting a misleading result.

6. Approach and methodology

The system will first establish simple forecasting baselines, then compare them with machine-learning models.

Candidate models include:

- Seasonal naive forecast.
- Moving average.
- Logistic regression.
- Random Forest.
- XGBoost or LightGBM.

The preferred initial model is XGBoost or LightGBM because the problem involves structured time-series and tabular features.

Possible model inputs include:

- Cases one to four weeks earlier.
- Recent case growth.
- Moving averages.
- Week of the year.
- Historical seasonal average.
- Rainfall one to three weeks earlier.
- Temperature.
- Neighbouring-area case trends.

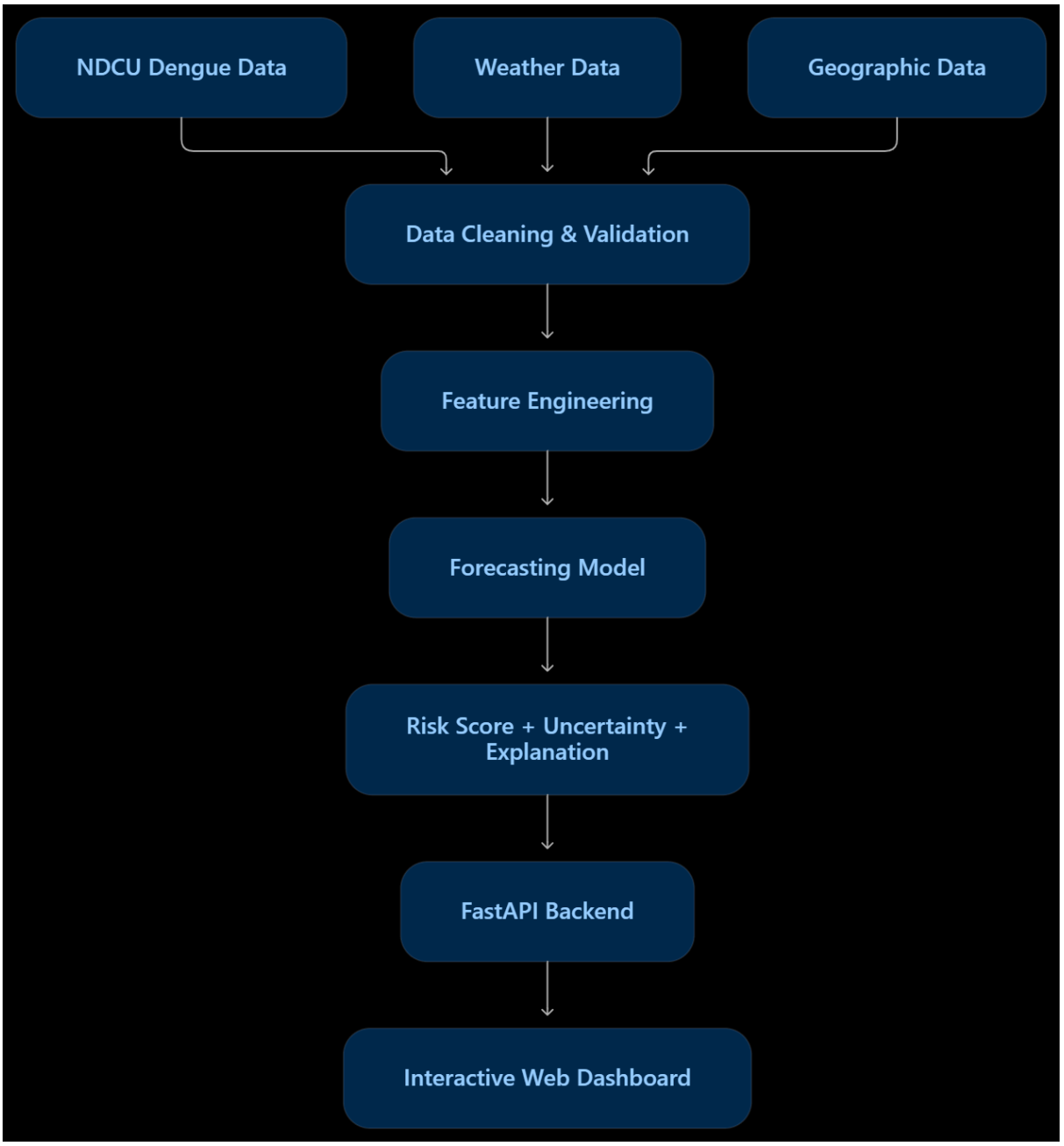
The team will only use more complex deep-learning methods if they provide measurable improvement over simpler models.

Where possible, SHAP will be used to explain the major factors influencing each prediction.

7. System design

7.1 Architecture

The system will follow this pipeline:



The system will also maintain information about data freshness and missing values.

7.2 Technical stack

Planned technologies include:

- Python.

- Pandas or Polars.
- Scikit-learn.
- XGBoost or LightGBM.
- SHAP.
- FastAPI.
- React/Next.js or Streamlit.
- Leaflet for maps.
- PostgreSQL or SQLite.
- GitHub and Docker.

Streamlit will be used as a fallback if frontend development takes too much time.

7.3 Interface and interaction

The main interface will be a map showing predicted dengue risk. Users will be able to click an area and see:

- Current cases.
- Predicted next-week cases.
- Hotspot probability.
- Seasonal expectation.
- Main contributing factors.
- Confidence.
- Data freshness.

The dashboard will also include a historical backtesting view to demonstrate how the model would have performed on previous outbreaks.

8. Implementation plan

20–22 October

- Validate dengue data.
- Build extraction pipeline.
- Clean historical dataset.
- Connect weather data.

23–25 October

- Build baseline models.
- Create lagged and seasonal features.
- Train XGBoost/LightGBM models.

26–27 October

- Perform historical backtesting.

- Measure lead time, recall, and precision.
- Add model explanations.

28–29 October

- Build API and dashboard.
- Add geographic visualisation.
- Add risk categories.

30–31 October

- Add uncertainty and fail-safe logic.
- Test deployment.
- Prepare documentation and demo.

1 November

- Final testing.
- Technical report.
- Repository.
- Two-page summary.
- Demo video.

If time becomes limited, optional features such as deep learning, LLM explanations, and advanced animations will be removed first.

9. Evaluation plan

The main goal is not simply high model accuracy. The system must identify hotspots **before they become obvious from current case counts**. Evaluation metrics will include:

Recall: How many future hotspots were identified in advance?

Precision: How many warnings actually became hotspots?

Lead time: How early was the warning generated?

Top-K recall: If an officer can only investigate the top 10 areas, how many true future hotspots appear in those 10?

Forecast error: Difference between predicted and actual case counts.

Calibration: Whether predicted probabilities correspond to real outcomes.

The model will be compared against:

- Last week's case count.

- Four-week moving average.
- Seasonal historical average.
- Ranking areas by current cases.

Historical backtesting will ensure that no future data is used during prediction. The approach will be considered unsuccessful if it does not consistently outperform these simple baselines.

10. Risks and contingencies

1. **Incomplete MOH-level data**
   Use district-level forecasting if needed.
2. **NDCU report format changes**
   Use modular extraction scripts and cached validated data.
3. **Weather API unavailable**
   Use recent cached data and reduce confidence.
4. **Model performs no better than baseline**
   Simplify the model and report the result honestly.
5. **Too many false alarms**
   Tune thresholds and probability calibration.
6. **Data becomes stale**
   Warn the user or withhold the prediction.
7. **Frontend development takes too long**
   Switch to Streamlit.
8. **Team member becomes unavailable**
   Keep tasks modular and documented in GitHub.

11. References

1. National Dengue Control Unit, Sri Lanka.
   https://www.dengue.health.gov.lk/
2. National Dengue Control Unit — Weekly Reports.
   https://www.dengue.health.gov.lk/weekly-report/
3. World Health Organization Sri Lanka. National Dengue Review 2026.
   https://www.who.int/srilanka/
4. Sri Lanka Department of Census and Statistics.
   https://www.statistics.gov.lk/
5. Open-Meteo Weather API.
   https://open-meteo.com/
6. Scikit-learn documentation.
   https://scikit-learn.org/
7. XGBoost documentation.
   https://xgboost.readthedocs.io/
8. LightGBM documentation.
   https://lightgbm.readthedocs.io/

9. SHAP documentation.
   https://shap.readthedocs.io/

12. Team details



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